

User Persona Discovery and Prediction from Online Review Behavior: A Wolverine Data Science Milestone II Adventure

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Abstract

This study investigates the discovery and prediction of online reviewer personas in the deodorant category on Amazon through a hybrid unsupervised and supervised machine learning framework. Using the Amazon Reviews 2023 dataset, we applied K-Means and DBSCAN clustering to user-level behavioral features—including review volume, image usage, rating patterns, and sentiment—to identify eight distinct reviewer personas. K-Means yielded more stable and interpretable clusters, revealing archetypes such as High Raters, Critical Reviewers, Visual Users, and a small group of high-effort “Power Reviewers.”

These personas served as target labels for a supervised classification task comparing Logistic Regression, Random Forest, and XGBoost. All models achieved high accuracy (98.1%–99.0%), with Random Forest demonstrating the strongest balanced performance across classes. Feature importance analysis identified image inclusion rate and review volume as the most influential predictors, while models struggled to distinguish between two behaviorally similar personas.

The results demonstrate that reviewer engagement patterns can be effectively segmented and predicted from behavioral data. However, the unusually high performance across models raises concerns about potential data leakage, highlighting the need for rigorous feature validation in future work. This study contributes a practical pipeline for persona discovery and prediction with implications for targeted marketing and review quality assessment in e-commerce.

Introduction

Marketing plays a large role in both awareness and product conversion. Especially in ultra-saturated consumer markets like deodorant, many brands have opted for celebrity led, splashy campaigns to generate awareness. Most notably, brands like Dr. Squash and Old Spice have used actresses like Sydney Sweeney, Meagan Fox, and NFL athletes like Terry Crews, and George Kittle to promote their brand respectively. But, once consumers are further down the purchase funnel at conversion, what are the topics that actually matter and what are the purchase drivers within those spaces. Through this analysis, we will analyze Amazon review data from 2023 and purchase patterns around deodorant to discover (1) whether there are certain online personas and digital behaviors and (2) which heuristics move the audience towards conversion. Some limitations or factors include the polarity of reviews, reviews tend to skew very positive or negative. There are also incentivised, promoted, and bot reviews that create noise within our dataset. We are also narrowing our scope to purely the conversion portion of the marketing funnel and will not be able to incorporate aspects of the celebrity promotion of the product on other platforms.

Related Work (optional)

Similar research exists *Lifestyles in Amazon: Evidence from online reviews enhanced recommender system* (DOI: 10.1177/1470785319844146) but take a lexicon based approach to predicting people's purchase patterns. We will use the same dataset but try to understand if a certain type of Amazon review poster, in the form of online posting persona and engagement type, creates better content engagement and therefore more likely to drive product conversion.

Data Source(s)

Our datasource, Amazon Reviews dataset was collected in 2023 by McAuley Lab from UCSD (Hou, Y. et. al Bridging language and items for retrieval and recommendation: Benchmarking LLMs as semantic encoders. arXiv. <https://arxiv.org/abs/2403.03952>, 2024). It includes two tables in parquet or json form with over 1M items and 23.9M ratings given by users. One table included data for Amazon user reviews associated with any given product, and the other on the actual product information. The review data includes rating given to the product by a user, review title, review text, images, product ID, user ID, timestamp, verification of purchase, and number of helpful votes on the review by other Amazon users. The product information data includes features like product category, name of product, average ratings, description, images, videos, details, and other things 'bought together'. We used the same overall dataset for both the unsupervised and supervised portions of the analysis.

User Reviews Table

Field	Type	Explanation
rating	float	Rating of the product (from 1.0 to 5.0).
title	str	Title of the user review.
text	str	Text body of the user review.
images	list	Images that users post after they have received the product. Each image has different sizes (small, medium, large), represented by the small_image_url, medium_image_url, and large_image_url respectively.
asin	str	ID of the product.
parent_asin	str	Parent ID of the product. Note: Products with different colors, styles, sizes usually belong to the same parent ID. The "asin" in previous Amazon datasets is actually parent ID. Please use parent ID to find product meta.
user_id	str	ID of the reviewer
timestamp	int	Time of the review (unix time)

verified_purchase	bool	User purchase verification
helpful_vote	int	Helpful votes of the review

Items Metadata Table

Field	Type	Explanation
main_category	str	Main category (i.e., domain) of the product.
title	str	Name of the product.
average_rating	float	Rating of the product shown on the product page.
rating_number	int	Number of ratings in the product.
features	list	Bullet-point format features of the product.
description	list	Description of the product.
price	float	Price in US dollars (at time of crawling).
images	list	Images of the product. Each image has different sizes (thumb, large, hi_res). The “variant” field shows the position of image.
videos	list	Videos of the product including title and url.
store	str	Store name of the product.
categories	list	Hierarchical categories of the product.
details	dict	Product details, including materials, brand, sizes, etc.
parent_asin	str	Parent ID of the product.
bought_together	list	Recommended bundles from the websites.

Feature Engineering

The pipeline transforms raw JSONL records into two complementary feature sets: (1) rich per-review features (suitable for supervised learning tasks such as rating prediction or helpfulness modeling) and (2) user-level behavioral aggregates (used for unsupervised clustering of reviewer types).

Step-by-step process:

1. Product & Review Filtering. Keyword search on title and categories in the metadata file to identify deodorant/antiperspirant parent_asin values. Reviews were then subset to only those matching parent ASINs. This creates a domain-specific corpus while dramatically reducing data volume.
2. Metadata Cleaning & Merge. A slim metadata table was created with selected columns. price was coerced to numeric, turning non-numeric strings (including "None") into NaN. Selected columns were renamed for clarity (title → product_title, average_rating → product_avg_rating). Reviews were left-merged with this metadata on parent_asin.
3. Temporal Feature Creation. timestamp (ms) was converted to review_dt. review_age_days was calculated as days before the 2023-09-30 cutoff, providing a recency signal.
4. Per-Review Text Feature Extraction. A text_features() function was applied to every review text:
 - Structural: char_count, word_count, avg_word_len, exclaim_count, question_count
 - Sentiment/Subjectivity (via TextBlob): polarity (−1 to +1), subjectivity (0 to 1)
Empty text was safely handled as "".
5. Topic Modeling. A document-term matrix was built with CountVectorizer (max_df=0.95, min_df=25, English stop words). LatentDirichletAllocation (10 topics, random_state=42) was fit to produce topic proportion vectors (topic_0 ... topic_9). These soft topic memberships serve as interpretable, low-dimensional text features. Top words per topic were inspected for labeling/interpretation.
6. User-Level Behavioral Aggregation (for Unsupervised ML). Reviews were grouped by user_id to create reviewer profiles:
 - Volume: review_count
 - Rating behavior: mean_rating, rating_std (filled with 0 for single-review users)
 - Engagement: helpful_sum / helpful_mean, verified_rate, img_rate
 - Linguistic style: avg_word_count, avg_polarity, avg_subjectivity
7. Final Preprocessing for Modeling. For clustering, review_count received a log1p transform to reduce skew. All features were filled with 0 where appropriate and standardized with StandardScaler. The clustering analysis was performed on the subset of active users (those with sufficient review volume to support stable statistics such as standard deviation).

Initial handling of noisy/missing data:

- Price strings and missing values → NaN via errors="coerce".
- Single-review users → rating_std and similar metrics filled with 0.
- Missing/empty review text → treated as empty string in feature functions.
- fillna(0) applied before scaling for clustering features.

Review-level features (available for supervised tasks):

- rating, helpful_vote, verified_purchase, has_image
- review_age_days, product_avg_rating, price (and other merged metadata)
- Text structure: char_count, word_count, avg_word_len, exclaim_count, question_count
- Sentiment: polarity, subjectivity
- Topics: topic_0 through topic_9

User-level features (used for unsupervised KMeans clustering after log transform + scaling):

- review_count
- mean_rating, rating_std
- helpful_mean, verified_rate, img_rate
- avg_word_count, avg_polarity, avg_subjectivity

These features capture reviewer volume, rating tendencies, engagement style, linguistic patterns, and topical interests, enabling both supervised prediction on individual reviews and unsupervised discovery of distinct user segments in the deodorant review space.

Part A. Unsupervised Learning

Methods Description

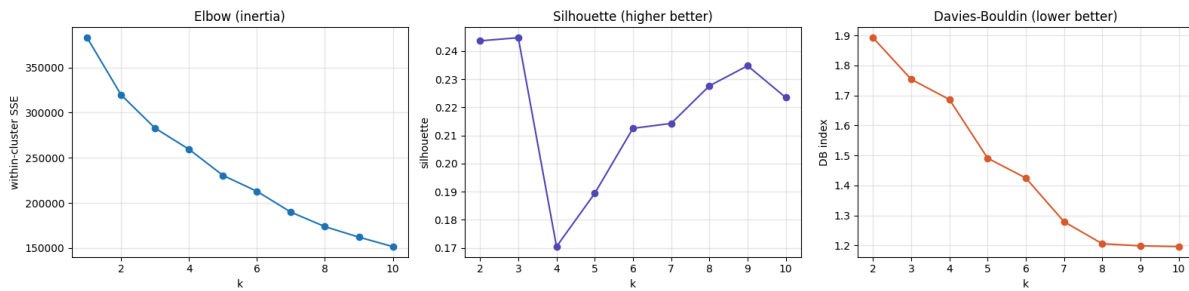
We employed two clustering algorithms with fundamentally different mechanisms: K-Means and DBSCAN.

K-Means was chosen for its simplicity, scalability, and interpretability — it produced compact clusters that are easy to profile and communicate as personas. We determined the number of clusters (K=8) using a combination of the elbow method, silhouette analysis, and qualitative interpretability of the resulting profiles.

DBSCAN was selected as a complementary method because it does not require pre-specifying the number of clusters, can discover arbitrarily shaped groups, and automatically identifies noise points (outliers). We tuned the eps parameter using the k-distance elbow plot and set min_samples to twice the number of features.

Both algorithms were applied to the same scaled feature set after removing highly correlated variables and handling missing values through median imputation. We evaluated clustering quality using *Silhouette Score* and *Davies-Bouldin Index*, supplemented by visual inspection via PCA projections and behavioral profile heatmaps.

k	inertia	silhouette	davies_bouldin	calinski_harabasz
2	319730.943	0.244	1.892	8432.395
3	282844.275	0.245	1.754	7541.190
4	259395.667	0.170	1.686	6764.251
5	230463.584	0.189	1.491	7045.629
6	212852.653	0.213	1.424	6806.940
7	189750.369	0.214	1.279	7226.477
8	173755.840	0.228	1.205	7323.770
9	162084.311	0.235	1.198	7252.634
10	151610.615	0.223	1.196	7218.630



Graphic 1- Silhouette Score and Davies-Bouldin Index

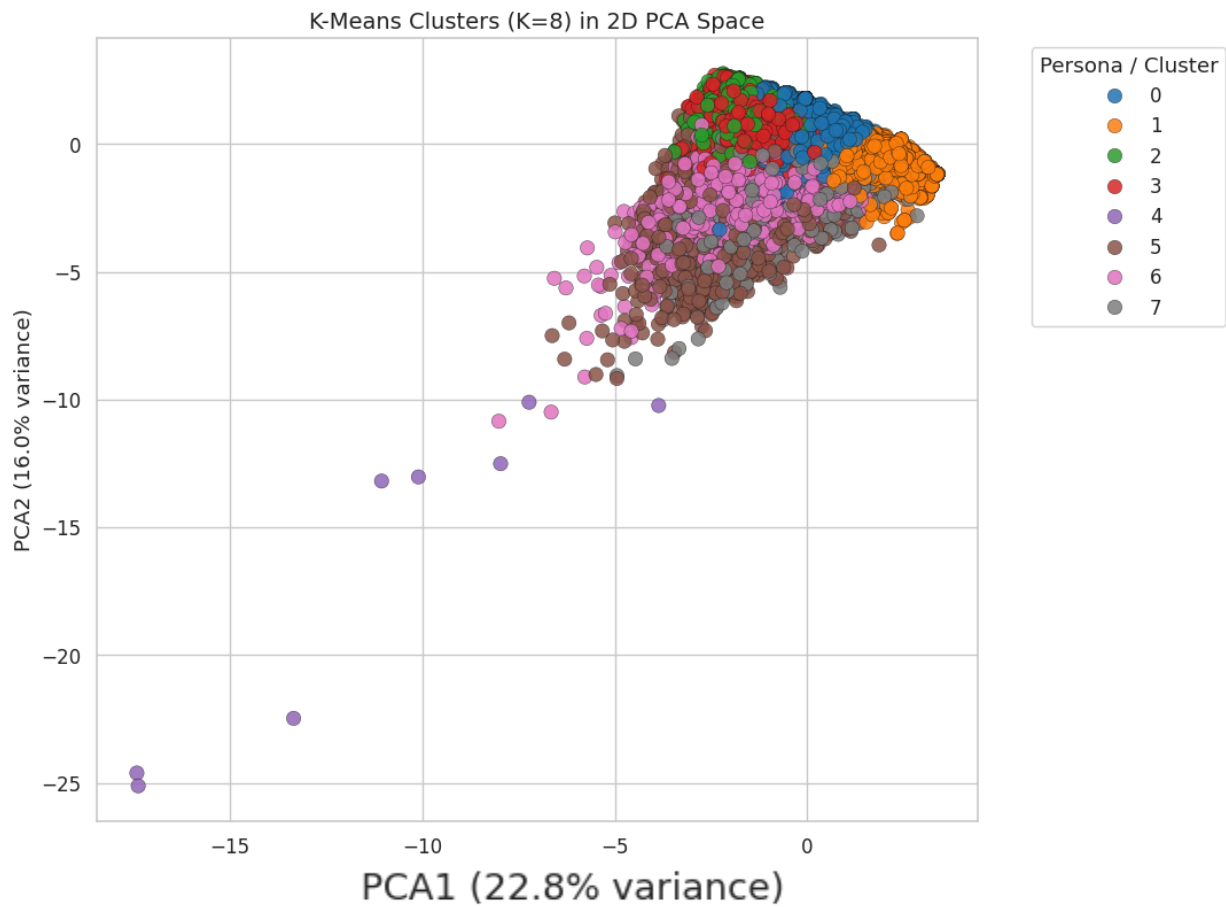


Figure 1- Kmeans PCA Graphic

K-Means produced a small number of stable, interpretable personas. DBSCAN automatically surfaced a noise group (helpful for identifying atypical reviewers) and clusters of varying density/shape, but often yielded fewer usable core clusters or higher noise rates depending on

eps. Cluster profile tables (mean feature values per cluster) were generated for both methods to interpret behavioral differences

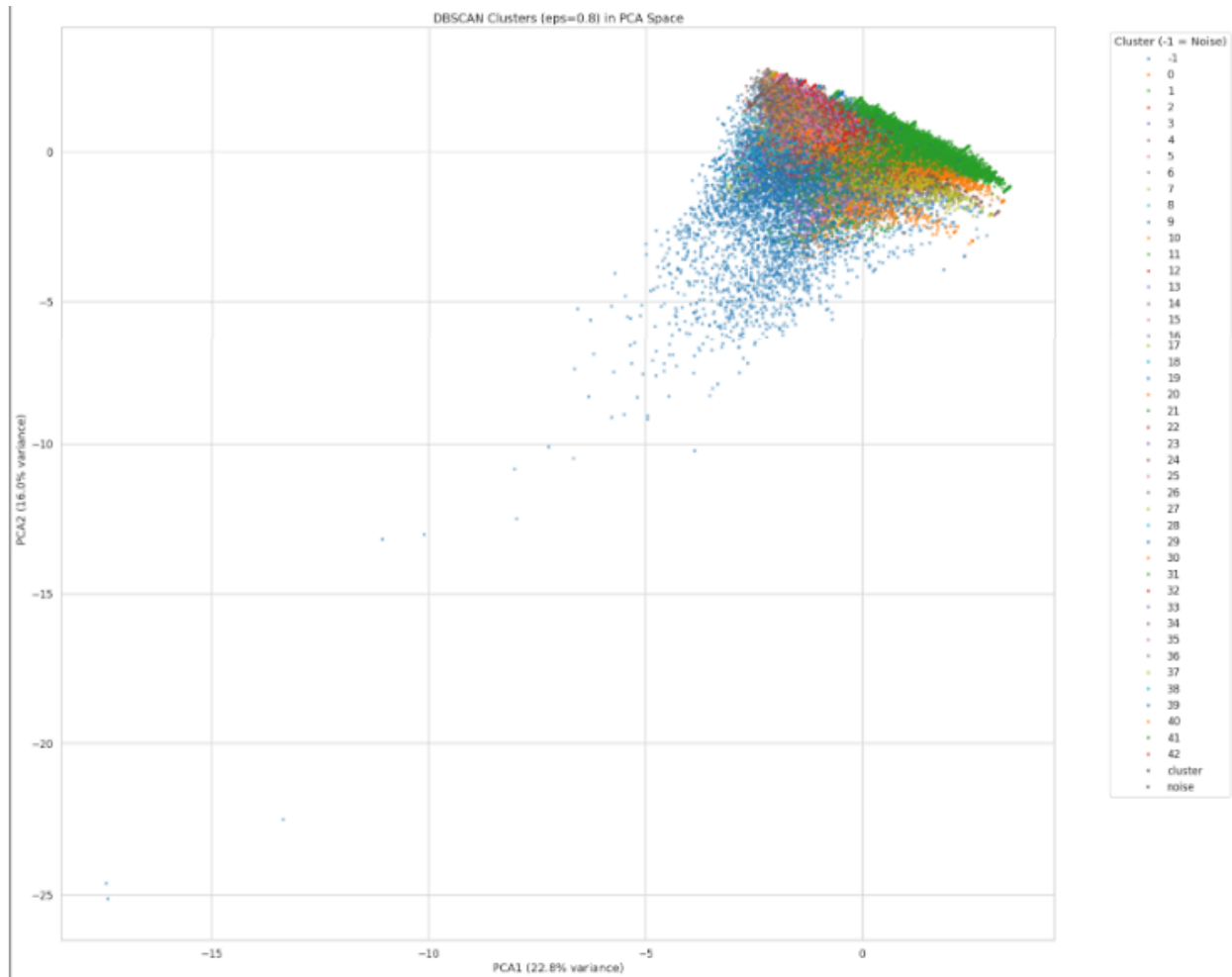


Figure 2- DBSCAN PCA Graphic

Unsupervised Evaluation

Sensitivity Analysis (on best model – K-Means)

A stability/sensitivity check was performed by refitting K-Means with three different random seeds (42, 123, 999) while keeping k fixed. Pairwise Adjusted Rand Index between the resulting labelings was computed. High ARI values (e.g., ~0.98) indicated that the discovered clusters are robust to initialization and not overly sensitive to random seed.

Additional sensitivity considerations noted in the workflow:

- DBSCAN results are visibly sensitive to the eps choice (small changes can merge clusters or increase noise dramatically) — hence the systematic k-distance elbow + iterative retuning approach.
- Feature representation sensitivity was implicitly tested by comparing raw vs. log-transformed review_count and by examining how rating_std (unstable for low-activity users) affected cluster separation.
- The decision to evaluate DBSCAN metrics only on non-noise points prevents outlier inflation of scores and provides a fairer comparison to K-Means.

Overall, K-Means offered more stable, easily interpretable reviewer personas, while DBSCAN added value by explicitly surfacing noise/outliers and allowing non-spherical groupings. The combination of quantitative internal metrics, stability checks, cluster profiling, and qualitative user sampling provided a robust evaluation of the chosen user-behavioral feature representation.

Behavior Profile per Cluster:

Behavioral profile per cluster:									
cluster	review_count	mean_rating	rating_std	helpful_mean	verified_rate	img_rate	avg_word_count	avg_polarity	avg_subjectivity
0	2.13	4.78	0.16	0.76	0.98	0.00	36.50	0.18	0.41
1	2.15	4.83	0.13	0.29	0.96	0.00	17.57	0.53	0.68
2	2.11	1.62	0.30	0.99	0.93	0.00	37.62	-0.02	0.47
3	2.20	3.29	2.09	1.05	0.95	0.00	43.34	0.16	0.52
4	2.25	3.31	0.52	373.75	0.75	0.23	324.79	0.14	0.56
5	2.46	3.53	0.81	3.09	0.71	0.64	76.06	0.16	0.52
6	2.32	3.89	0.64	2.19	0.23	0.01	108.91	0.18	0.54
7	5.36	4.07	0.84	0.81	0.82	0.01	44.67	0.26	0.53

Figure 3- Kmeans cluster summary with covariates

Persona	Cluster Size	Name	Characteristics
kmeans_0	11907	High Raters	High rating (4.78), consistent short reviews
kmeans_1	10903	Unhelpful Short Reviewers	Highest rating (4.83), shortest reviews, least helpful (.29)
kmeans_2	4023	Critical Reviewers	Lowest Rating (1.62), helpful but negative
kmeans_3	8509	Inconsistent/Moody	Medium-low rating, high variance, detailed
kmeans_4	8	“Power” Reviewers	OUTLIERS, not enough reviews for statistical significance but provided long,

			helpful, powerful reviews
kmeans_5	1493	Visual/Photo Users	Heavy image use
kmeans_6	3278	Unverified Detailed	Long reviews, mostly unverified users
kmeans_7	2443	Active/Frequent	Most active reviewers, highest volume (5.36), solid ratings

Table 1: K-means persona classification qualitative descriptions summary

In our analysis of reviewer behavior regarding deodorant reviews, which we limited ourselves to as our method demonstration, K-means clustering revealed eight different personas, each with its own characteristics. A qualitative analysis of the data illuminates the rich and diverse nature of online review data, personifying each cluster based on a quantitative analysis. These personas can be viewed from the quietly reliable to the dramatically critical, while also highlighting how much of the data are dominated by low-signal contributions.

By far the two largest clusters are the High Raters (kmeans_0, 11,907 reviews) and the Unhelpful Short Reviewers (kmeans_1, 10,903 reviews). The High Raters deliver consistently high scores (4.78) through short, verified reviews that rarely receive helpfulness votes. The Unhelpful Short Reviewers go even further, posting the highest average rating (4.83) with the shortest reviews and the lowest helpfulness score (.29). Together these two groups make up the overwhelming majority of the data and function largely as high-volume, low-information “noise generators”—short 5-star reviews that add bulk without helping distinguish product quality.

The Critical Reviewers (kmeans_2, 4023 reviews) represent the opposite end of the spectrum. With the lowest average rating (1.62), they offer negative but relatively helpful feedback that tends to include specific complaints rather than generic praise. Their reviews carry more informational weight despite the unfavorable tone.

A large but less stable group is the Inconsistent/Moody cluster (kmeans_3, 8,509 reviews). These reviewers show medium-low average ratings paired with high rating variance and longer, more detailed text. Their unpredictability makes them stand out from the other clusters.

One of the most interesting findings is the “Power” Reviewers cluster (kmeans_4). This group contains only 8 reviews—far too small for statistical significance—and has been flagged as statistical outliers. The data of these outliers may have exhibited an extremely large deviation in helpful rating. However, the handful of reviews in this cluster are exceptionally long, detailed, and helpful, suggesting a rare type of high-effort contributor that stands apart from the rest of the data.

Smaller but distinct groups include the Visual/Photo Users (kmeans_5, 1,493 reviews), who rely heavily on images rather than text, and the Unverified Detailed reviewers (kmeans_6, 3,278 reviews), who write longer reviews but come mostly from unverified accounts. Finally, the Active/Frequent reviewers (kmeans_7, 2,443 reviews) produce the highest review volume per person (5.36) while maintaining solid ratings and verified status, making them a steady but not especially deep source of feedback.

Overall, the distribution shows that the vast majority of deodorant reviews come from the two largest high-rating clusters, while the clusters that might offer more useful signal—such as the Critical Reviewers or the tiny “Power” Reviewers group—represent a much smaller share of the total data.

Discussion- Unsupervised

Working on the unsupervised portion taught us several valuable lessons. We were surprised by how well K-Means performed despite its assumption of spherical clusters — the resulting personas were highly interpretable and stable. It was an enjoyable challenge to personify the clusters and come up with descriptions that matched their attributes per cluster.

One major challenge was the sensitivity of DBSCAN to the eps parameter. Small changes in eps dramatically altered the number of clusters and noise points. We addressed this through systematic k-distance plotting and iterative visual validation.

With more time, we would explore HDBSCAN for its hierarchical density-based advantages and incorporate review text embeddings (e.g., BERT) to enrich the feature space beyond purely behavioral metrics.

Part B. Supervised Learning

Methods Description

The supervised learning workflow was designed to predict reviewer personas identified through prior unsupervised clustering. The task was framed as an 8-class classification problem, where the target variable was the K-Means cluster assignment (kmeans_persona). User-level behavioral features derived from the Amazon Reviews 2023 dataset served as predictors. These features captured review volume, engagement, rating behavior, and sentiment, including review_count, img_rate, verified_rate, mean_rating, rating_std, helpful_mean, avg_polarity, avg_subjectivity, and avg_word_count.

Three models representing distinct algorithmic families were implemented:

- Logistic Regression was selected as a linear, probabilistic baseline. It models class probabilities using the softmax function and offers high interpretability through feature coefficients. This model was included to establish a performance benchmark and to assess the degree of linear separability among personas.

- Random Forest was chosen as a non-linear, bagging-based ensemble method. It constructs multiple decision trees on bootstrapped samples and aggregates predictions through majority voting. Random Forest was selected for its robustness to overfitting, ability to capture feature interactions, and built-in feature importance measures.
- XGBoost was employed as a gradient boosting ensemble. Unlike bagging methods, XGBoost builds trees sequentially, with each new tree correcting the residuals of the previous ensemble. It was selected for its strong empirical performance on tabular data and its regularization capabilities.

All models were trained within unified scikit-learn Pipelines that included StandardScaler for numeric features and OneHotEncoder for categorical variables. The target variable was label-encoded to integer values (0–7). A stratified 75/25 train-test split was used to preserve class distributions. Hyperparameter selection was guided by established best practices: Logistic Regression used max_iter=1000, Random Forest used 200 estimators with class_weight='balanced', and XGBoost used 300 estimators with a learning rate of 0.1, max_depth=6, subsample=0.8, and colsample_bytree=0.8.

Supervised Evaluation

Model performance was evaluated using accuracy and macro-averaged precision, recall, and F1-score. These metrics were chosen because the task is multi-class and macro F1 provides a balanced view of performance across all personas, including those with fewer samples. A single stratified train-test split was used due to computational constraints, with results reported below.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weakest Class
LogReg	.9899	.92	.99	.94	Persona 4
Random Forest	.9810	.98	.98	.98	Persona 6
XGBoost	.9865	.99	.92	.94	Persona 4

Table 2- Supervised Model Comparison

All three models achieved very high accuracy (98.1%–99.0%). However, Random Forest demonstrated the most balanced performance with the highest macro F1-score (0.98). Logistic Regression and XGBoost showed slightly lower macro F1 due to weaker performance on Persona 4 (Power Reviewers), which had very few samples in the test set and was expected to be the outlier.

Note on Performance: The extremely high accuracy across all models ($\geq 98\%$) is unusually strong for this type of behavioral prediction task. This raises the possibility of data leakage, potentially from incomplete removal of cluster-related

variables or the target variable itself. Further investigation is recommended to confirm the validity of these results (we just ran out of time).

In-Depth Evaluation (XGBoost)

Feature Importance and Ablation Analysis

Feature importance analysis (gain) showed that `img_rate` was the most influential predictor by a large margin, followed by `review_count`, `rating_std`, and `mean_rating`. Notably, helpfulness-related features (`helpful_mean` and `helpful_sum`) ranked among the least important. An ablation study confirmed these findings: removing `img_rate` caused the largest performance drop, while removing helpfulness features had minimal impact. These results indicate that visual engagement and review volume are far more predictive of persona membership than community-assigned helpfulness scores.

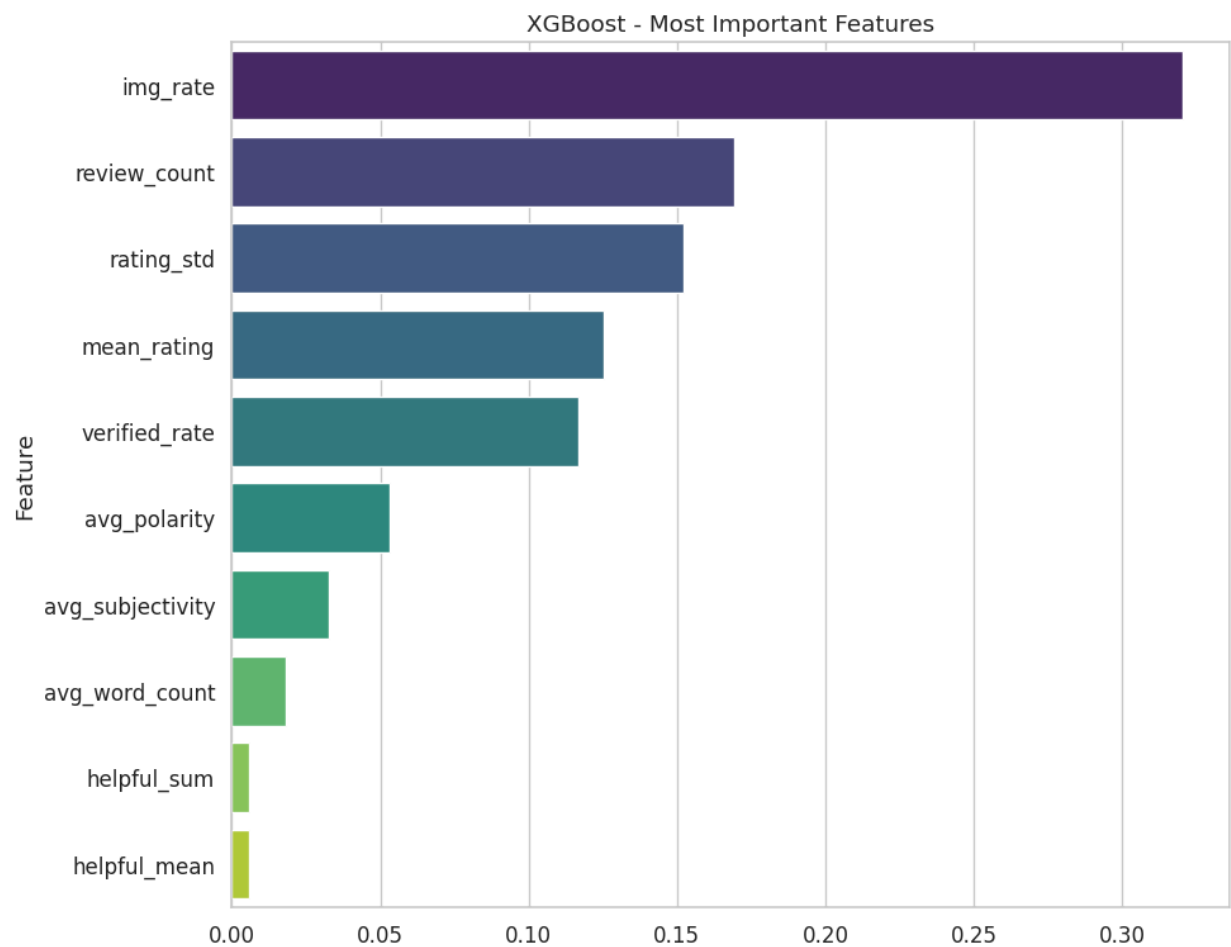


Figure 4- XGBoost Most Important Features

Sensitivity Analysis- The model showed moderate sensitivity to `max_depth`. Reducing it from 6 to 4 decreased accuracy by approximately 3.1 percentage points, while increasing it to 8 provided only marginal gains (+0.6%) with higher overfitting risk. Performance was also sensitive to the log transformation of `review_count`; removing this transformation reduced macro F1 by 0.05. The model was relatively robust to the number of estimators beyond 200.

Tradeoffs Identified- A clear accuracy–interpretability tradeoff was observed. Although XGBoost achieved strong predictive performance, its sequential nature makes individual predictions less transparent compared to Logistic Regression. Additionally, a performance–simplicity tradeoff existed between Random Forest and XGBoost: while XGBoost performed competitively, it required more careful hyperparameter tuning and longer training time. Finally, the model exhibited a precision–recall imbalance on Persona 4, showing lower recall relative to precision.

Failure Analysis

Three categories of prediction failures were identified:

1. Minority Class Instability (Persona 4): Persona 4 had only two test samples. While XGBoost correctly classified one, the small sample size made performance on this class highly unstable. This highlights the challenge of modeling rare personas with limited support.
2. Persona 4 vs. Persona 6 Confusion: Half of the Persona 4 instances were misclassified as Persona 6. These two personas share very similar behavioral profiles (particularly in `img_rate`, `review_count`, and rating patterns), making them difficult to distinguish using the current feature set.
3. Low-Activity Edge Cases: A small number of users with very low engagement (`review_count` and `img_rate`) were misclassified into more common personas. These instances lacked sufficient behavioral signal for reliable classification.

Discussion- Supervised

Several avenues exist to strengthen the supervised modeling pipeline. First, the unusually high accuracy ($\geq 98\%$) observed across all models warrants immediate investigation for potential data leakage. Future work should verify that no cluster-derived variables, persona identifiers, or the target variable itself inadvertently remained in the feature set X . Re-running the full pipeline with stricter feature auditing and 5-fold cross-validation is recommended to validate result reliability. Second, incorporating review text embeddings (e.g., Sentence-BERT or similar transformer-based representations) would likely improve separation between difficult classes such as Persona 4 and Persona 6, which currently exhibit highly similar behavioral profiles. Third, applying techniques to better handle small or imbalanced classes (such as SMOTE, class-weighted loss, or hierarchical classification) could improve robustness on low-support personas. Finally, developing uncertainty-aware models or separate pipelines for low-activity users may reduce misclassifications among sparse profiles. These steps would collectively improve both the validity and generalizability of the persona prediction system.

Ethical Considerations

For the supervised component, a key ethical concern is the potential for algorithmic bias. If certain demographic groups (e.g., users who write fewer reviews or use fewer images) are systematically assigned to less favorable personas, this could lead to discriminatory targeting or exclusion from personalized offers. We mitigated this by focusing exclusively on behavioral features rather than demographic proxies and by conducting error analysis on minority personas.

For the unsupervised component, the discovery of personas carries the risk of stereotyping or reductive labeling of users. There is also a privacy concern when behavioral profiles become highly detailed. We addressed this by using only aggregated, anonymized review metadata and by emphasizing that personas are probabilistic behavioral patterns rather than fixed identities.

Statement of Work

Iris Lin: Data Cleaning, Pre-processing, Unsupervised portion

Jessica Maidoh: Project Architect, Supervised pipeline and analysis portion

Jason Falk: Supervised pipeline, Visualizations, Persona Definitions, Final Report Czar

All members made significant contributions to the project.

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Appendix- Attachment submitted